1.2 Probability Space

$(\Omega, \mathcal{F})$

Probability measure

Let Ω be a sample space and $\mathcal{F}$ a σ-algebra on Ω. The map $\mathbb{P} : \underbrace{\mathcal{F}}_{\text{input}} \to \underbrace{[0,1]}_{\text{range}}$ is called a probability or probability measure such that the following axioms (Kolmogorov's axioms) are satisfied,

(by definition)

1. $\mathbb{P}(A) \geq 0$ *for all* $A \in \mathcal{F}$,
2. $\mathbb{P}(\Omega) = 1$,
3. *If* $A_1, A_2, \cdots \in \mathcal{F}$ *are countable and disjoint sets* $(A_i \cap A_j = \emptyset, i \neq j)$, *then*

$$\mathbb{P}\left(\bigcup_{n>0} A_n\right) = \sum_{n>0} \mathbb{P}(A_n).$$

Example Flip a biased coin.

$\Omega = \{H, T\}$, $\mathcal{F} = \{\emptyset, \{H\}, \{T\}, \overset{\Omega}{\{H,T\}}\}$

Probability: $\mathbb{P} : \mathcal{F} \to [0,1]$ s.t. $\mathbb{P}(\Omega) = 1$

$\mathbb{P}(H) = p$

$\mathbb{P}(T) = 1-p$

$\mathbb{P}(\emptyset) = 0$

$\mathbb{P}(\emptyset \cup \Omega) = \mathbb{P}(\emptyset) + \mathbb{P}(\Omega)$

$\Rightarrow \cancel{\mathbb{P}(\Omega)} = \mathbb{P}(\emptyset) + \cancel{\mathbb{P}(\Omega)}$

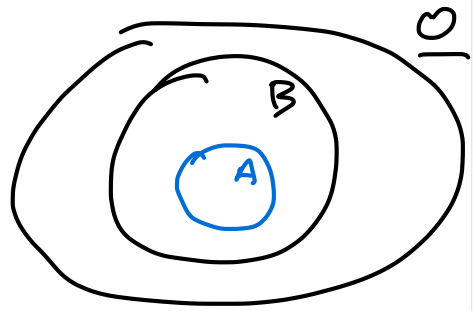
$\mathbb{P}(\emptyset) = 0$

Theorems

4) $\mathbb{P}(\emptyset) = 0$

5) if $A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$ (monotonicity)

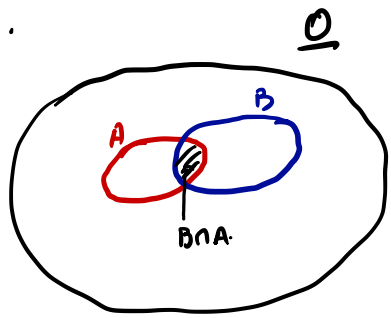
$$\mathbb{P}(B) = \mathbb{P}(A \cup (B \cap A))$$
$$= \mathbb{P}(A) + \underbrace{\mathbb{P}(B \cap A)}_{\geq 0} \geq \mathbb{P}(A)$$



6) $\mathbb{P}(A) \leq 1$, for any $A \in \mathcal{F}$

7) $\mathbb{P}(B \cap A^c) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

$$\mathbb{P}(B) = \mathbb{P}((B \cap A) \cup (B \cap A^c))$$
$$= \mathbb{P}(B \cap A) + \mathbb{P}(B \cap A^c)$$



Example

Continuous Ω

$\mathbb{P}(A) = (\text{Area in } A)/(\text{Area } \Omega)$

$\mathbb{P}(\Omega) = 1$

? $\Omega = \bigcup_n C_n \Rightarrow \mathbb{P}(\Omega) = \mathbb{P}(C_1) + \mathbb{P}(C_2) + \cdots = 0$.

Why this is wrong?

Probability Space

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called probability space.

Steps

0) Define the experiment.

1) Define the sample space

2) Construct σ-algebra $\mathcal{F}$

3) For each event A (= union of sample points) where $A \in \mathcal{F}$, use axiom 3 to find $\mathbb{P}[A]$

Example

0) Flip a coin (expériment)

1) $\Omega = \{H, T\}$ (sample space)

2) $\mathcal{F} = \{\phi, \{H\}, \{T\}, \Omega\}$ (σ-algebra)

3) $\mathbb{P} : \mathcal{F} \to [0,1]$

$\mathbb{P}(\{H\}) = 1/2$

$\mathbb{P}(\{T\}) = 1/2$

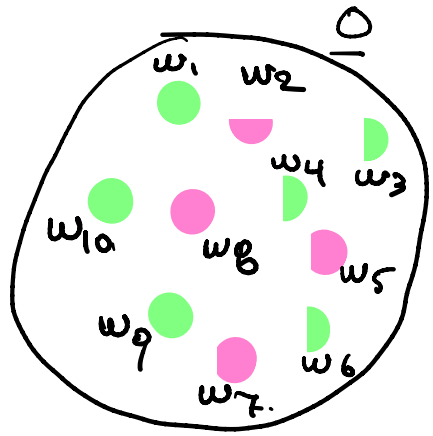
$\mathbb{P}(\Omega) = 1$

$\mathbb{P}(\phi) = 0$

(Probability measure)

Probability space: $\{\Omega, \mathcal{F}, \mathbb{P}\}$

Example



0) Experiment: Pick a sphere

1) Sample Space: $\Omega = \{\omega_i\}_{i=1}^{10}$

2) $\mathcal{F} = \{\phi, \Omega, G, M\}$

$G = \{\omega_1, \omega_3, \omega_4, \omega_6, \omega_9, \omega_{10}\}$

$M = \{\omega_2, \omega_5\ \omega_7\ \omega_8\}$

3) $\mathbb{P}(\omega_i) = \frac{1}{10}$ for each i

$\mathbb{P}(G) = \mathbb{P}(\omega_1) + \mathbb{P}(\omega_3) + \mathbb{P}(\omega_4) + \mathbb{P}(\omega_9) + \mathbb{P}(\omega_{10}) = \frac{6}{10}$

$\mathbb{P}(M) = \mathbb{P}(\omega_2) + \mathbb{P}(\omega_5) + \mathbb{P}(\omega_7) + \mathbb{P}(\omega_8) = 4/10$